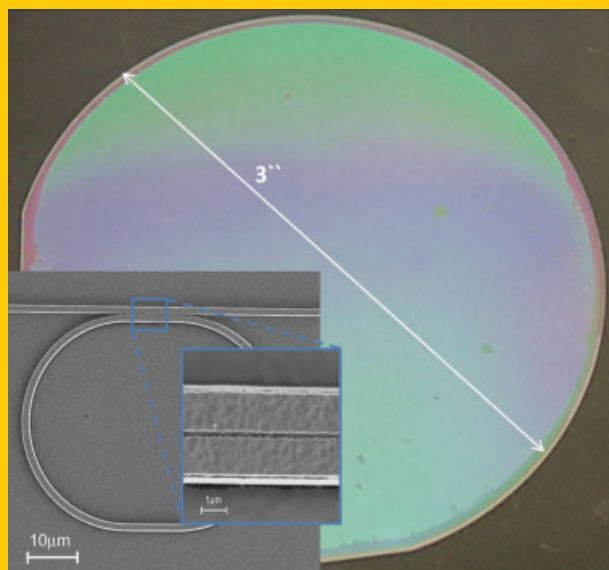


Abstract The state-of-the-art of high-refractive-index-contrast single-crystalline thin lithium niobate (LiNbO_3) films as a new platform for high-density integrated optics is reviewed. Sub-micrometer thin LiNbO_3 films are obtained by “ion-slicing”. They can be bonded by two different techniques to a low-index substrate to obtain “lithium niobate on insulator” (LNOI) even as wafer of 3" diameter. Different micro- and nano-structuring techniques have been used to successfully develop micro-photonic devices. To be specific, the fabrication and characterization of LNOI photonic wires with cross-section $< 1 \mu\text{m}^2$, periodically poled LNOI photonic wires for second harmonic generation, electro-optically tunable microring resonators, free standing microrings for hybrid integration, and photonic crystal structures are described.



Lithium niobate on insulator (LNOI) for micro-photonic devices

Gorazd Poberaj^{1,3,*}, Hui Hu^{2,4}, Wolfgang Sohler², and Peter Günter^{1,5}

1. Introduction

As optical components continue to replace electronics for optical signal processing applications, there is a growing impetus to integrate more photonic devices onto a single chip. Compared to electronics, photonic devices offer several advantages, such as large bandwidth operation, wavelength division multiplexing, and also absence of electro-magnetic interference. In the last few years, strong efforts have been made to develop silicon based photonic chips [1]. This field of research has profited considerably from the availability of large-size silicon-on-insulator (SOI) wafers and the advanced semiconductor technology providing solutions for micro- and nano-structuring of photonic devices. A high refractive index contrast between silicon and silicon-oxide is essential for the realization of highly integrated optics devices based on photonic wires [2–4] and photonic band gap structures [5]. SOI photonic wires can have a very small cross-section (even below $1 \mu\text{m}^2$) and also small bending radius ($\sim 10 \mu\text{m}$).

A corresponding technology for lithium niobate (LiNbO_3) is still in its infancy, though LiNbO_3 offers – in contrast to SOI – excellent electro-optic, acousto-optic, and nonlinear

optical properties. Moreover, it can be easily doped with rare-earth ions to get a laser active material [6]. Therefore, LiNbO_3 photonic wires will enable the development of a wide range of extremely compact, active integrated devices, including electro-optical modulators, tunable filters, nonlinear wavelength converters, and amplifiers and (tunable) lasers of different types. In particular, periodically poled LiNbO_3 (PPLN) photonic wires are ideal candidates for extremely efficient nonlinear optical devices for wavelength conversion and all-optical signal processing.

The key step towards realization of high-index-contrast LiNbO_3 waveguides is to produce single-crystalline LiNbO_3 thin films, which can be structured and embedded in low index dielectric materials. In this endeavor, many techniques have been studied in the past, such as chemical vapor deposition [7], RF sputtering [8], molecular beam epitaxy [9], sol-gel [10], and pulsed laser deposition [11]. However, all these techniques have difficulty in producing high crystalline quality materials. In addition, the use of substrates with required properties is limited, in particular for epitaxial growth due to lattice matching constraints. More recently, crystal ion slicing in combination with wafer bonding techniques has emerged as a very promising technique for fab-

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rication of single-crystalline ferroelectric thin films. This method generally known under the name of “Smart Cut” was originally discovered and applied for the fabrication of SOI wafers [12]. It uses high-dose implantations of H^+ and/or He^+ ions (dose $D \sim 10^{16}$ – $10^{17}/\text{cm}^2$) for cleaving films from a bulk material. Since the process was disclosed in 1995 [13], several variants of this method have also been studied for the exfoliation and transfer of different ferroelectric thin films.

Using crystal ion slicing technology, free-standing LiNbO_3 films with areas of several cm^2 were successfully fabricated and devices were demonstrated [14]. Crystal ion slicing was combined with direct bonding to fabricate LiNbO_3 films on SiO_2 of an area of $\sim 1 \text{ cm}^2$ [15]. Using benzocyclobutene (BCB) as cladding layer, LiNbO_3 films of a size of several cm^2 were realized and electro-optical tunable micro-ring resonators were reported [16]. Recently, a full wafer technology to fabricate single crystalline LiNbO_3 layers of sub-micrometer thickness on a vacuum deposited SiO_2 layer on a LiNbO_3 substrate has been developed [17]. In contrast to the BCB bonded material, annealing at much higher temperatures is possible. High temperature annealing can minimize ion-implantation induced material damage and helps to recover electro- and nonlinear optical properties [18, 19]. Moreover, high temperature annealing contributes significantly to a reduction of the surface roughness of LiNbO_3 ridge waveguides.

This paper is structured as follows. Section 2 describes the fabrication of ion-sliced lithium niobate thin films. In particular, it reviews two different techniques for bonding submicrometer ion-sliced LiNbO_3 films on low-index substrates to obtain “lithium niobate on insulator” (LNOI) samples and even wafers. Section 3 focuses on micro- and nanostructuring techniques which have been developed for prototyping of high density integrated optical devices in LNOI. The first low-loss and even periodically poled photonic wires in LNOI are presented in Sect. 4. Section 5 reviews the first realization of electro-optically tunable microring resonators. Section 6 demonstrates the suitability of LNOI as a platform for photonic crystal structures. Section 7 concludes the review and gives prospects for future developments of high-density integrated optics devices based on LNOI.

2. Fabrication of LNOI

The first step to fabricate LNOI consists of “ion-slicing” a thin film of LiNbO_3 from a single crystalline LiNbO_3 sample. A high-dose implantation with He^+ ions is used to define a “cleavage plane” within the crystal (Sect. 2.1).

Before cleaving is done by a special annealing process, the ion-implanted crystal is bonded to a substrate. Two different techniques for bonding on low-index substrates have been developed to obtain high quality LNOI samples and even full wafers.

In the first approach, bonding is performed using BCB on a LiNbO_3 substrate, which turned out to be a very reliable and reproducible way for the fabrication of large-area (several cm^2) LNOI samples. Using a polymer adhesive has the

advantage of substantially relaxed requirements for flatness, roughness, and cleanliness of the bonded surfaces. Prior to bonding, the LiNbO_3 substrate can also be coated with a thin metal layer serving as bottom electrode in electro-optically active devices (Sect. 2.2).

In the second approach, crystal bonding to SiO_2 -coated LiNbO_3 was used to fabricate LNOI. In contrast to the BCB bonded material, annealing at much higher temperatures is possible. This is important to recover electro- and nonlinear optical properties of the cleaved LiNbO_3 film. Moreover, high temperature annealing significantly reduces the surface roughness of ion beam etched LiNbO_3 (Sect. 2.3).

Crystal bonding is a very versatile process. It was also used to fabricate various sandwich structures with a thin metal electrode either underneath the SiO_2 layer or underneath the LiNbO_3 film; such structures should enable electric field induced poling and electro-optic control of light propagation. Also LNOI samples with MgO-doped LiNbO_3 core layers have been developed (Sect. 2.4).

2.1. Crystal ion slicing

The crystal ion slicing technique is based on high dose ion implantation and epitaxial lift-off. Several studies [14, 15, 20–23] have shown that He^+ implantation is suitable for slicing of LiNbO_3 . In the implanted layer, He forms gaseous bubbles, also referred to as blisters. Basically, thin film exfoliation or lift-off from an implanted LiNbO_3 crystal can be achieved either by wet-etching [20, 24] or thermal treatment [15, 21]. The reported methods differ also in bonding techniques used for the transfer of brittle exfoliated thin films onto supporting substrates. Bonding is one of the most critical steps in fabrication of large-area thin films.

The wet-etching technique exploits selective etching: the ion implanted damaged layer (induced at the ion penetration depth) can be etched by HF with much higher rates than undamaged, crystalline LiNbO_3 . An etching selectivity as high as 10 000 has been reported for an optimized process [25]. The final bonding of the detached film on the support substrate can be performed after etching and annealing, so that any extensive thermal cycling of the film/substrate composite can be avoided. However, this method is limited to films of many micrometer thickness. They vary continually and gradually in thickness, being thicker in the center and thinner towards the edges, where the exposure to HF was longer [24].

In order to fabricate uniform sub-micrometer thin LiNbO_3 films, thermally induced exfoliation has been used. Thermal treatment induces a growth of microcavities in the damaged layer, leading to micro-cracks and finally to a detachment of the implanted layer. A controlled split-off or exfoliation is possible only after proper bonding of the implanted sample onto a supporting substrate. Since the thermal treatment requires elevated temperatures ($> 220^\circ\text{C}$) for the exfoliation of implanted LiNbO_3 crystals, the different thermal expansion of the bonded materials is a very critical issue. A thermal expansion mismatch can induce a large

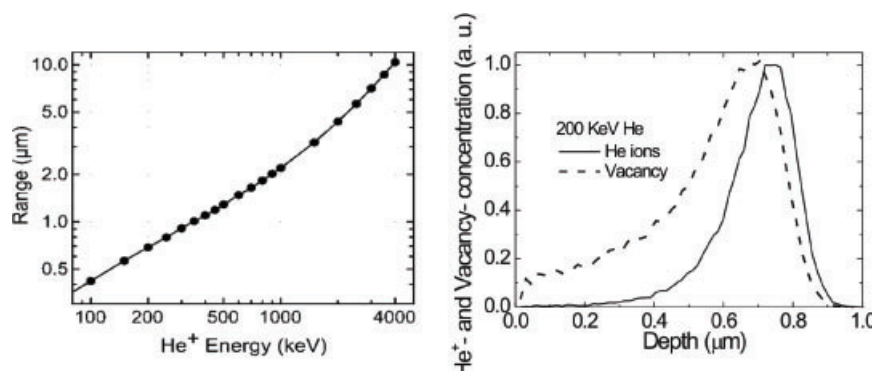


Figure 1 Left: Calculated range of He⁺ ions implanted into lithium niobate as function of their energy [29]. Right: He⁺- and vacancy-distributions calculated by SRIM.

mechanical stress, causing cracks and voids in the transferred film. In the following subsections, two techniques are described, which have successfully overcome this problem.

Since the thickness of an ion-sliced film is determined by the implantation depth of the implanted He⁺ ions, the required film thickness for a particular application can be adjusted by the implantation energy. Figure 1 shows an average penetration depth (range) of implanted He⁺ ions in LiNbO₃ as function of their energy according to “Stopping and Range of Ions in Matter” (SRIM) calculations [26]. An implantation energy of ~ 200 keV is suitable for ion-slicing of sub-micrometer thin layers. The obtained planar LiNbO₃ waveguides on a low index cladding support single transverse modes (in vertical direction) at 1550 nm wavelength.

2.2. BCB bonding

In this approach, developed at ETH Zurich [16], the bonding is performed using an adhesive polymer benzocyclobutene (BCB) [27, 28], which turned out to be a very reliable and reproducible way for fabrication of large-area (several cm²) LiNbO₃ thin films. Using a polymer adhesive has the advantage of substantially relaxed requirements for the flatness, roughness, and cleanliness of the bonded surfaces (donor and acceptor wafer). Optically, BCB has excellent transparency in the visible and infrared region, and as a cladding layer provides a suitable optical confinement owing to its low refractive index ($n \approx 1.55$).

The whole fabrication procedure [16, 29] is schematically shown in Fig. 2. A Z- or X-cut congruent LiNbO₃ wafer with a diameter of 3" and a thickness of 0.5 mm is implanted with He⁺ ions with an energy of 195 keV and a dose of $\sim 4.5 \times 10^{16} \text{ cm}^{-2}$. The implanted wafer is then cut into smaller samples, which are cleaned using standard RCA1 solution and bonded to LiNbO₃ substrates by means of BCB. Prior to the bonding, the substrates can also be coated with a thin metal layer (e.g., 50 nm of Cr) serving as a bottom electrode in electro-optically active devices. After the bonding, the sample is gradually heated up. The thermal treatment has a threefold role: in the slow-ramp heating phase the bonding strength is increased, at around 220 °C the implanted layer splits-off from the donor wafer as a single-crystalline film, and this is then annealed at 300 °C for several hours to partially recover implantation-induced crystal defects. Finally,

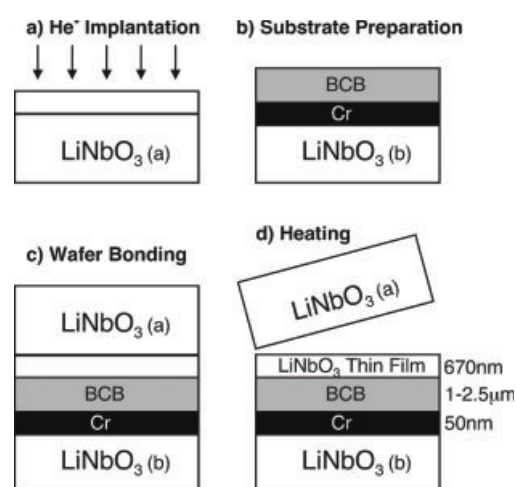


Figure 2 Schematic of the LiNbO₃ thin film fabrication method: (a) He⁺ implantation of LiNbO₃ donor wafer ($E = 195 \text{ keV}$, $D \sim 4.5 \times 10^{16} \text{ cm}^{-2}$); (b) preparation of LiNbO₃ receptor wafer (substrate); deposition of Cr electrode and spin coating of an adhesive polymer film (BCB); (c) wafer bonding; (d) slow-ramp thermal treatment: strong bonding, LiNbO₃ film split-off, and annealing.

the surface of the thin film is smoothened by Ar⁺ ion etching or mechanical polishing. The later can reduce the surface roughness below 1 nm rms.

Fig. 3a shows a photograph of a Z-cut LiNbO₃ thin film with dimensions of 12 mm × 10 mm × 600 nm bonded on a Cr-coated LiNbO₃ substrate. Such samples show good mechanical stability and chemical inertness needed for further

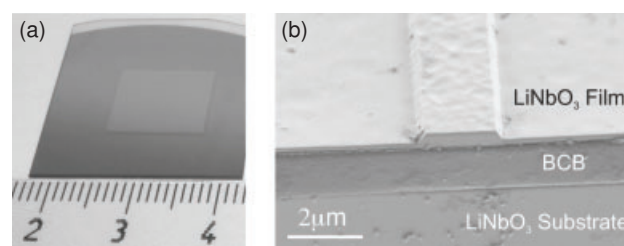


Figure 3 (a) Photograph of an ion-sliced LiNbO₃ thin film (12 mm × 10 mm × 600 nm) bonded on a Cr-coated LiNbO₃ substrate. (b) Polished edge of a LiNbO₃ thin film with a structured ridge waveguide for end-fire light coupling.

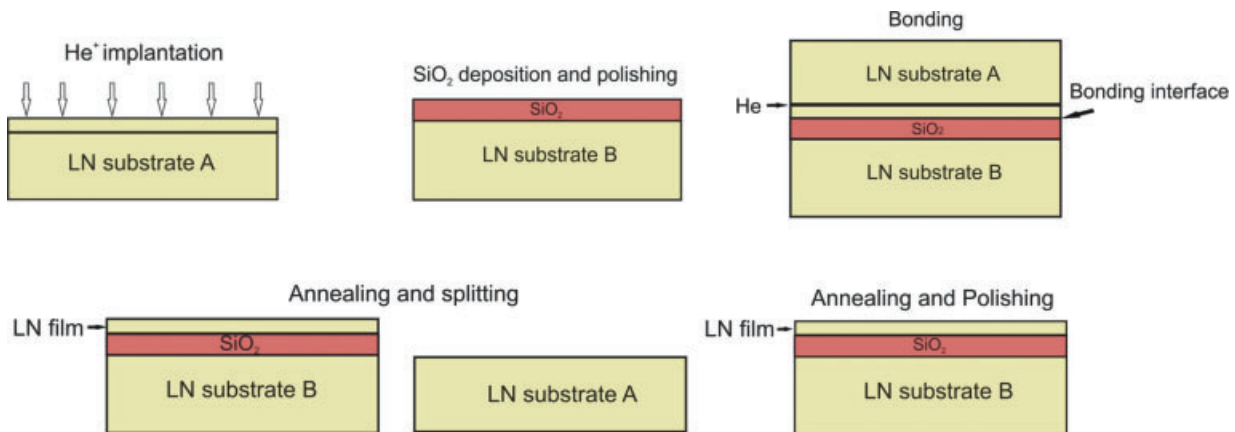


Figure 4 (online color at: www.lpr-journal.org) Fabrication scheme of LNOI: a “smart cut” single crystalline LiNbO_3 (LN) film of sub-micrometer thickness is crystal-bonded to a SiO_2 / LiNbO_3 (LN) substrate.

processing steps. Figure 3b shows a polished edge of a 200-nm thick LiNbO_3 film with a 600-nm high ridge waveguide for end-fire light coupling.

Single-crystalline LiNbO_3 films of this type were used as a platform for the first realization of electro-optically tunable microring resonators [16] which will be presented in Sect. 5. More recently, single-crystalline LiNbO_3 photonic crystal slabs were also fabricated and their optical properties were characterized [30] as described in Sect. 6.

2.3. Crystal bonding to SiO_2

To fabricate a LNOI structure, another choice of insulator is SiO_2 between the ion-sliced LiNbO_3 film and the substrate. In contrast to the BCB bonded material, annealing at much higher temperatures becomes possible. High temperature annealing can minimize ion-implantation induced material damage and helps to recover electro- and nonlinear optical properties. Moreover, it contributes significantly to a reduction of the surface roughness of LiNbO_3 ridge guides [31]. To avoid a thermal coefficient mismatch, the substrate should be LiNbO_3 bulk material as well.

A SiO_2 layer of high density is deposited on the LiNbO_3 substrate using plasma-enhanced chemical vapor deposition (PECVD), which also provides a very good adhesion. Moreover, the stress between SiO_2 and the LiNbO_3 substrate can be controlled by the parameters of the PECVD process. The SiO_2 surface roughness should be reduced by polishing to < 1 nm in order to enable a high bonding strength of ion-sliced LiNbO_3 film and SiO_2 layer.

Such directly bonded layer structures have been pioneered by the group of Günter [15] on smaller substrates. Recently, the Paderborn group succeeded in the fabrication of LNOI wafers of 3" diameter [32] as described below.

2.4. Wafer scale fabrication of LNOI

In the following, the full-wafer process to fabricate LNOI is briefly described [17,32]. Also different sandwich structures

with a thin metal electrode either underneath the SiO_2 layer or underneath the LiNbO_3 film have been fabricated. Such structures should enable electric field induced poling and electro-optic control of light propagation. Even LNOI with a $\text{MgO}:\text{LiNbO}_3$ film on SiO_2 on a LiNbO_3 substrate has been demonstrated.

Full-wafer fabrication

The full wafer fabrication process is schematically shown in Fig. 4. At first, a Z-cut LiNbO_3 wafer of 3" diameter is implanted by He-ions with a dose of 4×10^{16} ions/ cm^2 forming an amorphous layer at a depth dependent on the implantation energy (see Fig. 1). Another Z-cut LiNbO_3 handle sample is coated by a SiO_2 -layer of 1.3 μm thickness by plasma enhanced chemical vapor deposition (PECVD), and then annealed at 450 °C for 8 hours to drive off the gases trapped in the oxide layer. With a chemical mechanical polishing (CMP) process, the surface roughness is reduced from about 6 nm to 0.35 nm enabling direct wafer bonding (Fig. 5). The bonded pair of samples is then annealed (16 hrs. @ 165 °C + 6 hrs. @ 190 °C) to improve the bonding strength. By a further increase of the temperature (2 hrs. @ 228 °C), the sample splits along the He implanted layer. Afterwards, it is annealed at 450 °C for 8 hours, before the

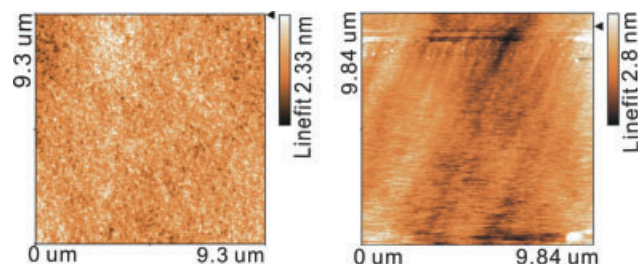


Figure 5 (online color at: www.lpr-journal.org) Left: AFM micrograph of a polished SiO_2 surface of RMS-roughness of 0.35 nm. Right: AFM micrograph of a polished LNOI surface of RMS-roughness of 0.5 nm.

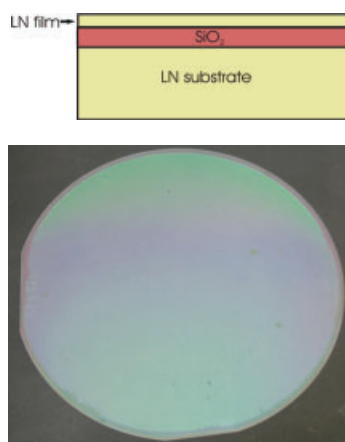


Figure 6 (online color at: www.lpr-journal.org) Schematic cross-section (top) and photograph (bottom) of a LNOI-wafer of 3" diameter after splitting, consisting of a 760-nm thick single crystalline LiNbO₃ (LN) film crystal bonded on a 1.3-μm thick SiO₂ layer deposited by PECVD on a LiNbO₃ (LN) substrate.

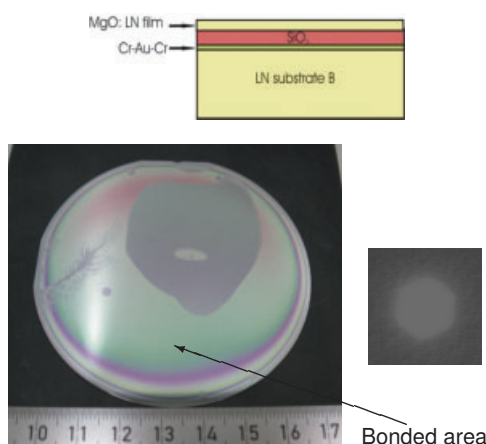


Figure 7 (online color at: www.lpr-journal.org) Schematic cross-section (top) and photograph (bottom) of a LNOI wafer of 3" diameter, consisting of a 1040-nm thick single crystalline MgO:LiNbO₃ (LN) film crystal bonded on a 1.3-μm thick SiO₂ layer deposited by PECVD on a thin (~40 nm) evaporated Cr-Au-Cr-electrode on a LiNbO₃ (LN) substrate. On the right: spontaneously formed micro-domain of inverted polarization after annealing at 450 °C for 5 hours. The size of the reversed domain is around 10 μm.



Figure 8 (online color at: www.lpr-journal.org) Schematic cross-section (top) and photograph (bottom) of a LNOI wafer of 3" diameter, consisting of a 1040-nm thick single crystalline LiNbO₃ (LN) film crystal bonded on a 20-nm thick evaporated Ti electrode on a 1.3-μm thick SiO₂ layer deposited by PECVD on a LiNbO₃ (LN) substrate. The upper surface of the LiNbO₃ (LN) film is not yet polished after splitting.

LNOI-surface is polished to 0.5 nm roughness by another CMP-process (Fig. 5).

The CMP process and subsequent surface cleaning are of key importance to achieve the required high quality surfaces with lowest surface roughness and excellent flatness enabling direct wafer bonding. A scratch-free polished surface is obtained by filtering the polishing slurry. RCA cleaning is used to remove the particle contamination on the wafers.

LiNbO₃ film / SiO₂ / LiNbO₃ substrate

Fig. 6 presents a 3 inch LNOI wafer. The good homogeneity of the interference colors underlines the homogeneity of the fabrication process. A thickness variation within 6% is measured by a step profiler.

MgO:LiNbO₃ film / SiO₂ / Cr-Au-Cr / LiNbO₃ substrate

Fig. 7 shows the result of experiments to fabricate a LNOI wafer of 3 inch diameter with a MgO-doped LiNbO₃ thin film on top and a Cr-Au-Cr metal electrode underneath the SiO₂ layer. The idea was to take advantage of the lower coercive field of MgO:LiNbO₃ and to facilitate (light assisted) electric field poling via a second electrode on top of the LNOI surface. However, corresponding experiments failed up to now; it seems that the required conductivity of the SiO₂ layer is too low to allow the flow of the poling current. On the other hand, some spontaneously generated micro-domains could be observed after the final annealing and polishing processes (see Fig. 7).

LiNbO₃ film / Ti / SiO₂ / LiNbO₃ substrate

Also sandwich structures with a thin Ti-film as poling electrode on top of the SiO₂ layer have been developed (see Fig. 8). They are currently investigated to get electric field poling via a second electrode on top of the LNOI surface. Poling voltages of a few tens of Volts should be sufficient to overcome the coercive field.

A metal layer of Ti has been chosen to allow a thermal oxidation after poling. We expect that the resulting TiO₂ layer of high refractive index would only slightly modify the LNOI waveguide characteristics and would avoid ohmic losses for guided wave propagation. First experimental results are promising.

3. Micro- and Nanostructuring of LNOI

Due to their high index contrast of $\Delta n \approx 0.7$ ($n_{\text{LiNbO}_3} \approx 2.14$, $n_{\text{SiO}_2} \approx 1.44$), LNOI samples are strongly guiding planar waveguides even with sub-micrometer thickness of the core layer. As such they are ideally suited for the fabrication of LNOI photonic wires and micro-optical devices. However, their fabrication requires advanced micro- and nanostructuring methods including high resolution and flexible photolithography, chemical and ion beam etching techniques.

Two different types of optical lithography have been used to prepare photoresist masks enabling subsequent etching of the LiNbO₃ layer of LNOI; they are presented in Sect. 3.1. However, in contrast to SiO₂, LiNbO₃ is much harder to etch in such a way that smooth surfaces result,

a prerequisite for low loss waveguiding especially in optical channels of sub-micrometer dimensions. Chemical and ion beam etching techniques have been developed in the last years to fabricate high quality LiNbO_3 surfaces of low roughness [31, 33]. For the fabrication of LNOI photonic wires and micro-optical devices, inductively coupled plasma (ICP) etching was used resulting in high quality photonic wires (Sect. 3.2) and microring resonators (Sect. 3.3).

For sub-micrometer fabrication, focused ion beam (FIB) etching is a very precise and controllable method. It was used to drill photonic crystal structures into the LiNbO_3 layer of LNOI samples (Sect. 3.4).

For efficient quasi phase matched (QPM) nonlinear optical interactions in photonic wires, periodically poled (PP) structures are required. However, periodic poling of LNOI photonic wires proved to be a big challenge up to now. Therefore, an alternative strategy was used to develop first periodically poled LNOI samples, which were processed afterwards in the same way as mono-domain samples to fabricate periodically poled photonic wires (Sect. 3.5).

3.1. Standard optical and laser lithography

As shown in our previous studies [16, 17] (see also Sect. 4), LiNbO_3 channel waveguides embedded in BCB ($n = 1.55$) or crystal bonded on SiO_2 ($n = 1.45$) must be narrower than $1\ \mu\text{m}$ in order to achieve single mode waveguiding at a wavelength of $1550\ \text{nm}$. Patterning of sub-micrometer wide waveguides and coupling gaps requires advanced techniques such as electron-beam- or laser-lithography or at least an electron-beam written mask for standard UV photolithography. While electron-beam lithography systems have higher resolution, laser writers can be more suitable for fast prototyping of integrated optical devices.

The Paderborn group used an electron-beam written mask of high precision to define photoresist stripes of $1\text{--}7\ \mu\text{m}$ width and a thickness of $1.7\ \mu\text{m}$ serving as etch masks. Due to the etching ratio of LiNbO_3 and photoresist of about 1.0, this relatively large thickness was necessary to enable an etching depth up to $460\ \text{nm}$ by using inductively coupled plasma (ICP)-etching with Ar-ions (see also Sect. 3.2). Figure 9 schematically shows the cross-section of a LNOI sample with a photoresist mask before and after Ar-milling. A typical thickness of the LiNbO_3 layer was $730\ \text{nm}$ crystal-bonded to a $1.3\text{-}\mu\text{m}$ thick SiO_2 layer on a Z-cut LiNbO_3 substrate. The CMP-polished surface had a rms-roughness of $0.5\ \text{nm}$.

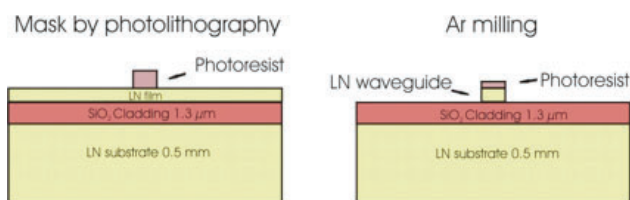


Figure 9 (online color at: www.lpr-journal.org) Schematic diagram of the cross-section of a LNOI sample (left) used to fabricate photonic wires by Ar-milling (right).

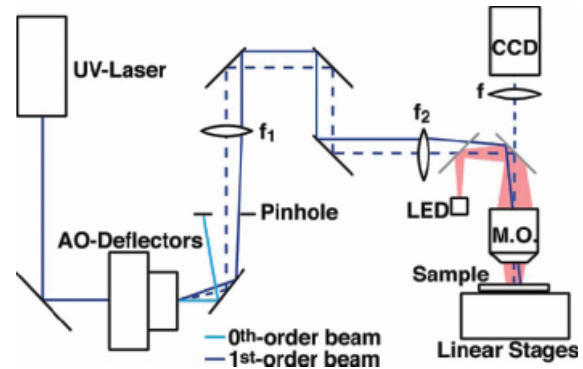


Figure 10 (online color at: www.lpr-journal.org) Scheme of the laser lithography system showing the main optical components and the laser beam path.

The Zürich group set up a compact laser lithography system operating at $375\ \text{nm}$ wavelength, which is schematically shown in Fig. 10. The system consists of a continuous-wave, frequency stabilized diode laser, two acousto-optic deflectors (AOD) in series at right angles to give full two-dimensional scanning, a beam steering controller for driving the AODs, a microscope objective for focusing the laser beam, and three (xyz) motorized linear stages to position a sample. A CCD camera is used for observing the sample surface and for the positioning of the laser beam. Using a microscope objective with a high numerical aperture ($\text{NA} > 0.75$), sub-micrometer wide patterns can be written by the focused laser beam. The use of the AODs substantially enhances writing speed and precision needed for demanding structures. A detailed description of the laser lithography system and its performance can be found in [34].

3.2. Argon-milling (ICP-etching) of photonic wires

Using an Oxford Plasmalab System 100 the Paderborn group fabricated series of channel guides with dimensions down to sub-micrometer width and height. Ar-milling was used for typically $60\ \text{min}$ with $100\ \text{W}$ RF-power inductively coupled into the plasma (ICP) and $70\ \text{W}$ RF power coupled to the sample table. The resulting etch depth was $460\ \text{nm}$. Figure 11 shows scanning electron micrographs (SEM) of a ridge guide of $1\ \mu\text{m}$ top width. The dark stripe underneath is the SiO_2 layer on the LiNbO_3 substrate. On both sides of the ridge, etched trenches can be observed, resulting from additional etching by ions reflected by the angled walls of the ridge. Their slope of $\sim 27^\circ$ (with respect to the vertical direction) is determined by the dependence of the etching rate on the incident angle of the ion beam. As the etched trenches apparently isolate the ridge, longer etching was not considered. However, it should be possible to completely remove the remaining LiNbO_3 layer by extending the etching time. Finally, the end faces of the sample were polished to enable efficient end-fire coupling of light. A typical sample length was $3\ \text{mm}$ [17]. Results of a detailed characterization of these photonic wires are presented in Sect. 4.

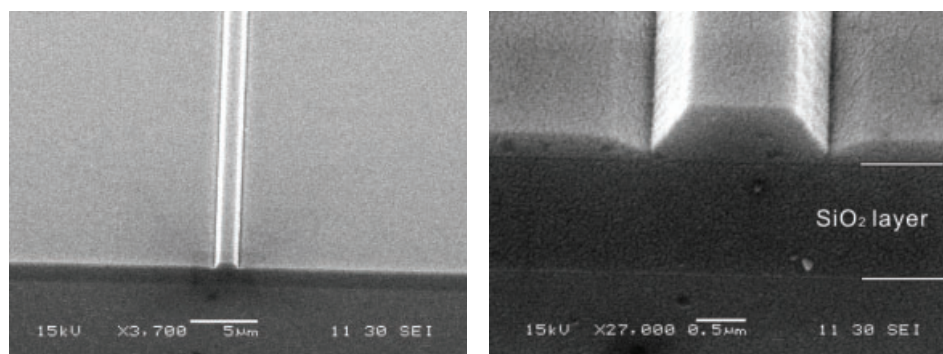


Figure 11 SEM micrographs of LNOI ridge guides (photonic wires) of sub-micrometer cross-section dimensions ($\sim 1 \mu\text{m}$ top width). The trenches on both sides of the ridges nearly reach the surface of the SiO_2 buffer layer.

3.3. Argon-milling (ICP-etching) of photonic microrings and couplers

Using an Oxford Plasmalab 80+ the Zürich group fabricated micro-ring resonators with coupling waveguides. Figure 12a shows an example of a microring with a coupling waveguide, structured by the laser lithography system in a negative-tone photoresist film span on a LiNbO_3 crystal. To achieve 200-nm narrow coupling gaps, a double lithography process was used. In the first step, coupling waveguides were written in the photoresist film by moving the sample with the translation stages, and then the photoresist was developed and hardened. In the second step, the whole lithography process was repeated with a new photoresist film and the rings were written by precise steering of the laser beam using the acousto-optic deflectors. Figure 12b shows an example of a microring with a coupling waveguide on a LiNbO_3 crystal surface after Ar^+ ion etching. The etching depth is about 400 nm.

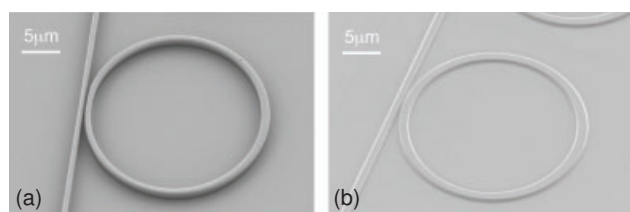


Figure 12 Scanning electron micrographs of a microring resonator with a coupling waveguide: (a) photoresist mask on a LiNbO_3 wafer written by the laser lithography system; (b) final structure transferred onto a LiNbO_3 crystal surface by Ar^+ ion ICP-etching (Ar-milling).

3.4. Focused ion beam milling

Fabrication of nanostructures such as arrays of cylindrical holes for photonic crystals requires even higher resolution. As shown in our previous study [30], focused ion beam (FIB) milling is very well suited for fast prototyping of high-quality cylindrical holes in ion-sliced LiNbO_3 thin films. In our experiments we used a Zeiss Nvision 40 device in combination with RAITH Elphy Quantum lithography system. Figure 13 shows a scanning electron micrograph

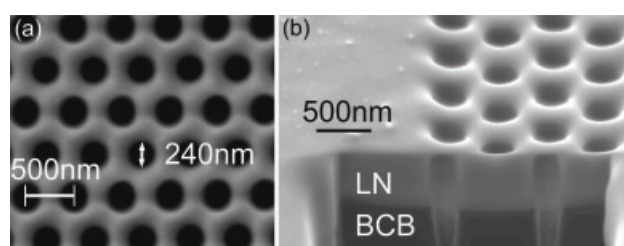


Figure 13 (a) Scanning electron micrograph of a triangular lattice of air holes in a Z-cut, 600-nm thick LiNbO_3 film milled by focused ion beam. The lattice constant is 500 nm and the air holes have a diameter of 240 nm. (b) Cross-section showing holes with a regular cylindrical shape in the LiNbO_3 thin film. The holes with a depth of 1 μm penetrate into the BCB underlayer.

of triangular lattice of air holes with a diameter of 240 nm and a lattice constant of 500 nm structured in a LiNbO_3 thin film. The holes were milled with a focused beam of Ga ions accelerated to 30 kV at a current of 300 pA. A single pass milling strategy was used which enables shorter processing times and is less prone to slight drifts of focused ion beam. In order to minimize unwanted charging effects during the ion beam milling, samples were covered with 5 to 10-nm thick chromium layer and electrically connected to the grounded sample holder. After the completion of the milling process, the chromium layer was removed by chemical etching. We note that the unwanted re-deposition effects in the milling process were minimized because of the relatively soft BCB underlayer, which facilitated the milling of cylindrical holes with vertical walls in LiNbO_3 thin films.

3.5. Periodically poled photonic wires

Up to now, direct electric field poling of LNOI samples or of photonic wires was not yet successful. We assume that small defects in the LiNbO_3 layer or in the LiNbO_3 channel lead to electrical breakdown before domain inversion is achieved. Therefore, we used an alternative strategy to develop periodically poled photonic wires as needed for QPM nonlinear optical interactions. Instead of starting with a single-domain LiNbO_3 substrate A (see Fig. 4) to fabricate a LNOI sample, a periodically poled substrate A was used and processed as described in Sect. 2.

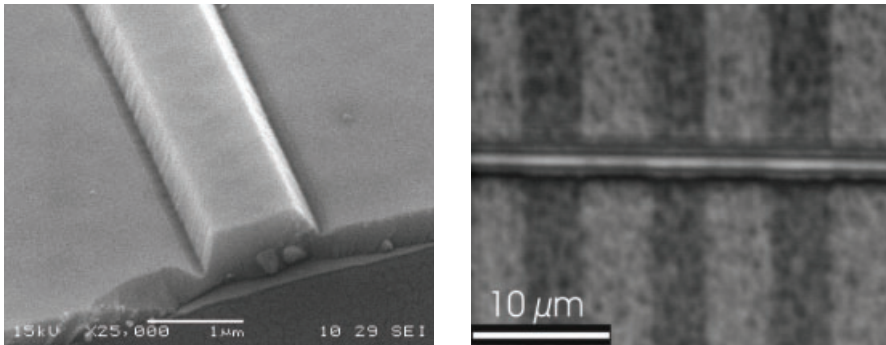


Figure 14 SEM micrograph (left) and optical micrograph (right) of a periodically poled LNOI photonic wire of 1 μm top width and 710 nm thickness. The periodicity of the ferroelectric domain grating is 9 μm .

The result was a periodically poled LNOI film. Also the definition and fabrication of channel guides using the PPLNOI film was done as discussed above for the single domain samples resulting in periodically poled LNOI photonic wires of 9 μm periodicity of the ferroelectric domain grating [35].

Figure 14 shows on the left a scanning electron microscope (SEM) micrograph of a PPLNOI photonic wire of 1 μm top width and 710 nm thickness. Again, on both sides of the ridge etched trenches can be observed, resulting from ions reflected by the angled walls of the ridge. As the domain structure was hardly visible in the SEM micrograph, also an optical micrograph was taken with a differential interference contrast microscope; it is shown on the right. The domain grating is clearly visible not only in the remaining LiNbO_3 film on both sides of the ridge, but also in the photonic wire itself.

4. Photonic wires

Photonic wires are important basic structures in integrated optics. They are channel waveguides of high index contrast and very small cross-section dimensions leading to strong light confinement. Due to their extremely small mode size, LiNbO_3 photonic wires are ideal candidates for efficient nonlinear optical devices. In particular, periodically poled structures would enable quasi phase matching (QPM) of second order nonlinear processes such as second harmonic generation (SHG), difference frequency generation (DFG), etc.

We have developed and carefully investigated several LiNbO_3 channels of sub-micrometer dimensions. Their fundamental optical properties were characterized around 1.55 μm wavelength. To be specific, their mode distributions were measured and compared with modeling results. The mode propagation losses were determined for different wire width and as function of the polarization using the resonator method [35]. A correct analysis required the knowledge of the waveguide end face reflectivity, which was evaluated for a number of parameters. Moreover, the resonator method allows determining the group index of the waveguides, which was measured in this way and also compared with modeling results [37]. Finally, (third order) quasi phase matched (QPM) second harmonic generation (SHG) was demonstrated in periodically poled LNOI photonic wires.

4.1. Coupling to photonic wires

A tunable laser diode was used to investigate the photonic wires. The light was coupled to the waveguides by a 60 $\times$ /0.8 objective with an estimated coupling efficiency of about 40% (30%) to the fundamental mode of the wire of 2 (1) μm top width. On the output side, a 100 $\times$ /0.9 objective magnified the near field distribution of the guided mode to form an image of maximum spatial resolution ($\sim 1.1 \mu\text{m}$) measured by a camera.

4.2. Mode distributions

The mode distributions were measured as function of the waveguide width and of the polarization, and compared with simulated distributions calculated with a full-vectorial finite-difference method. According to the simulation results, both structures support not only the fundamental, but at least also the second order mode at 1.55 μm wavelength. However, by a careful adjustment of the input laser beam, the fundamental mode could be selectively excited. Moreover, the remaining thin LiNbO_3 films of 270 nm thickness on both sides of the photonic wires formed a planar waveguide for qTE polarization.

Figure 15 shows the measured mode distribution (a) for quasi TE-polarization (qTE) together with a corresponding

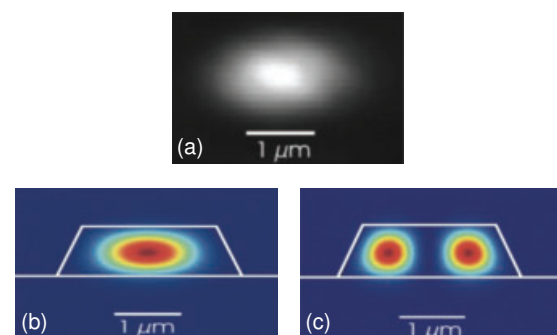


Figure 15 (online color at: www.lpr-journal.org) Measured (a) and calculated (b) intensity distribution of the fundamental mode in a photonic wire of 2 μm top width (qTE-polarization; $\lambda = 1.55 \mu\text{m}$). The intensity distribution of the simulated second order mode is shown in (c). The profile of the ridge, used for the simulations, is indicated (b,c).

modeling result (b) at the output of a 2- μm wide photonic wire. In view of the finite spatial resolution of the imaging system, the agreement is reasonable. The calculated second order mode is displayed in Fig. 15c.

Fig. 16 shows the measured (a) and calculated (b) distribution of the fundamental quasi TM mode (qTM) in a photonic wire of 1 μm top width. Again, in view of the finite spatial resolution of the imaging system, the agreement is reasonable. The intensity distribution of the simulated second order mode is shown in Fig. 16c.

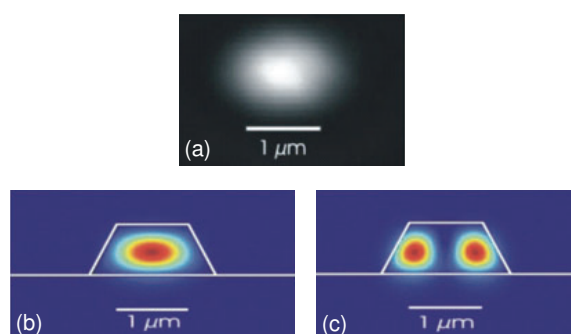


Figure 16 (online color at: www.lpr-journal.org) Measured (a) and calculated (b) intensity distribution of the fundamental mode in a channel waveguide of 1 μm top width (qTM-polarization; $\lambda = 1.55 \mu\text{m}$). The intensity distribution of the simulated second order mode is shown in (c). The profile of the ridge, used for the simulations, is indicated (b,c).

The cross-section of the fundamental mode of the 1 μm wide photonic wire is about $0.4 \mu\text{m}^2$ (product of FWHM in horizontal and vertical directions of the calculated distribution), more than one order of magnitude smaller than the mode size of about $16 \mu\text{m}^2$ in a conventional Ti-indiffused strip guide of 7 μm width. Therefore, such photonic wires are ideal candidates to develop nonlinear devices of high efficiency. Moreover, high bending radii become possible without inducing excess bending losses.

4.3. End face reflectivities

Due to the small waveguide cross-section, the modal reflectivity R (ratio of reflected mode power to the incident mode power) can no longer be approximated by the Fresnel result for plane waves of normal incidence at a LiNbO_3 / air

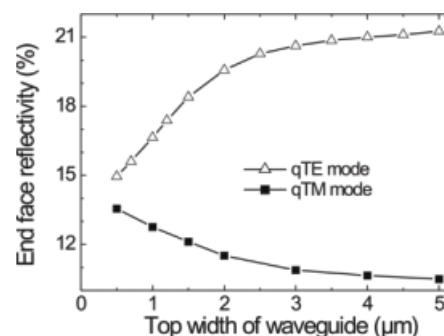


Figure 17 Calculated end face reflectivities of the fundamental modes of qTE and qTM polarization, respectively, versus the top width of LiNbO_3 photonic wires of 730 nm thickness.

interface ($\sim 14\%$) as usually done for weakly guiding structures. The actual value of R is needed to enable a correct evaluation of the propagation losses by using the resonator method [36] (see also next subsection). Therefore, R was calculated by a 3-dimensional finite difference time domain (FDTD) solver [38]. The result is presented in Fig. 17 as function of the top width of the ridge guides of 730 nm thickness. The slope of the ridges was kept constant assuming an angle of 27° with respect to the normal on the waveguide substrate. With increasing waveguide width, the end face reflectivity R approaches polarization dependent limits with a significantly higher reflectivity for qTE than for qTM modes. Due to the small waveguide thickness of 730 nm only, the reflectivity difference $\Delta R = R_{\text{qTE}} - R_{\text{qTM}}$ converges to $\sim 11\%$. This figure shrinks to $\sim 1\%$ with increasing thickness, determined by the birefringence of LiNbO_3 alone. With decreasing waveguide width (at 730 nm thickness), ΔR also shrinks to small figures, as qTE and qTM modes become nearly degenerate.

4.4. Propagation losses

The polished waveguide end faces of reflectivity R , as calculated above, form low-finesse waveguide cavities. By analyzing the contrast K of the cavity resonances, measured as function of the wavelength, the mode propagation losses were determined. This method is independent on the mode excitation efficiency [36].

Figure 18 shows two examples: on the left, the measured normalized transmission of a channel guide (photonic wire)

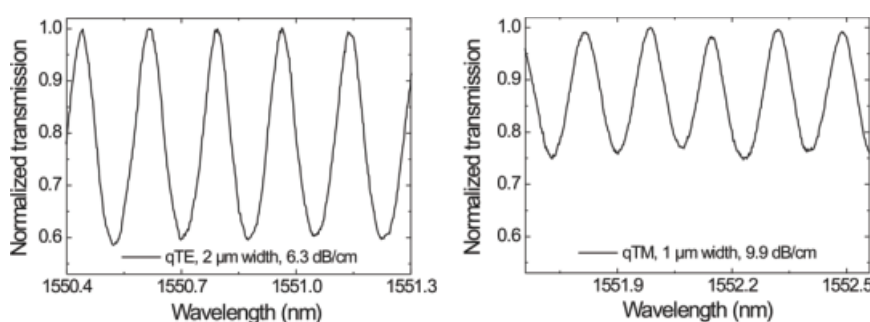


Figure 18 Normalized transmission of a channel guide of 2 (1) μm top width for qTE (qTM) polarization as function of the wavelength (left/right).

of 2 μm top width for qTE polarization as function of the wavelength is presented; on the right, the transmission of a channel guide of 1 μm top width for qTM polarization is shown. A qTE-propagation loss of 6.3 dB/cm was evaluated for the 2- μm wide channel; for qTM polarization, the loss is 7.5 dB/cm. In the 1- μm wide photonic wire the qTM propagation loss was determined to be 9.9 dB/cm; for qTE polarization, the loss is 12.9 dB/cm. The larger qTE losses in the narrower wire may result from some light coupling to the thin planar (single polarization) waveguides on both sides of the channel.

The losses very likely arise from scattering by the residual roughness of all four walls of the ridge. Losses due to a coupling of the propagating light across the SiO_2 layer to the substrate should be negligible (< 0.05 dB/cm), according to the calculations with a fully vectorial finite difference mode solver.

4.5. Effective and group indices

The waveguide mode dispersion is determined by the material dispersion of core and cladding materials and also by the waveguide dimensions [39]. Due to the high index contrast of the photonic wires and their small cross-section dimensions, the effective index of refraction n_{eff} varies considerably as function of the wavelength. It was calculated by the finite difference method for the fundamental qTE- and qTM-modes [40], respectively, and for several waveguide widths and wavelengths; some results are displayed in Fig. 19 together with the dispersion of bulk LiNbO_3 and SiO_2 .

Whereas the effective mode index n_{eff} is difficult to measure, the group index n_g could be derived in a small wavelength range from the measured transmission spectra of the waveguides (see Fig. 18): $n_g = \lambda^2 / (2 \cdot L \cdot \Delta\lambda)$; L is the waveguide length and $\Delta\lambda$ is the wavelength difference of neighboring resonances [39, 41, 42]. In addition, the group index was also calculated as $n_g = n_{\text{eff}} - \lambda \cdot (dn_{\text{eff}}/d\lambda)$; measured and calculated results are shown in Fig. 19 for the fundamental modes of qTE and qTM polarization, respectively.

For comparison, also the calculated group index of bulk LiNbO_3 is shown derived from a Sellmeier equation describing the dispersion of bulk LiNbO_3 [43].

The effective indices of refraction vary between the bulk indices of LiNbO_3 and SiO_2 ; the longer the wavelength and the smaller the waveguides' cross sections are, the lower the effective indices become due to a decreasing mode confinement. At shorter wavelengths, the mode is more and more confined in the LiNbO_3 core of the waveguide and n_{eff} approaches the (polarization dependent) bulk index of LiNbO_3 . The group indices for light propagation in the photonic wires are substantially larger than for light propagation in bulk LiNbO_3 , reflecting the influence of the small waveguide dimensions. They even allow adjusting a group index dispersion close to zero in the 1.5 μm wavelength range, which is of interest for optical communication devices.

4.6. Quasi phase matched (QPM) SHG in periodically poled photonic wires

Due to the modified dispersion in the nonlinear LNOI photonic wires, also their phase matching properties for nonlinear interactions are correspondingly dependent on the waveguide dimensions [44]. They were theoretically investigated for second harmonic generation (SHG) taking also periodically poled domain gratings into account to get quasi phase matched (QPM) interactions. For periodically poled (PP)LNOI photonic wires, the calculations showed that type I QPM SHG for a fundamental wave of 1064 nm wavelength should be possible in a wire of 1 μm top width and 730 nm thickness with a domain grating of 3 μm periodicity. As such a small periodicity has not yet been fabricated, the third order QPM SHG was considered via the corresponding Fourier component of the 9 μm grating already developed (see Fig. 14). This process should be possible at the expense of a reduced efficiency. Figure 20 (left) shows the simulated normalized SHG characteristics (for 3 μm periodicity) for different lengths of the grating; the shorter the nonlinear

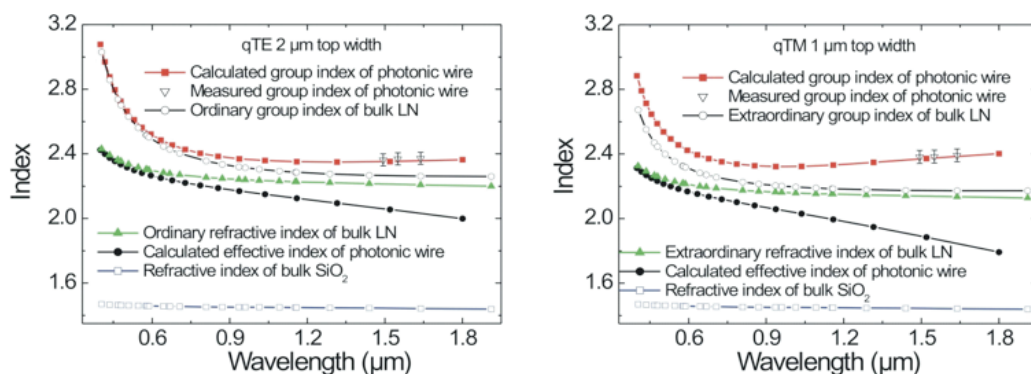


Figure 19 (online color at: www.lpr-journal.org) Calculated effective and group indices for the fundamental modes of qTE and qTM polarization in photonic wires of 730 nm thickness and of 2 μm and 1 μm top width, respectively, versus the wavelength. The measured group indices of photonic wires, the calculated group indices of bulk LiNbO_3 (LN), and the refractive indices of bulk LiNbO_3 and SiO_2 are shown as well for comparison.

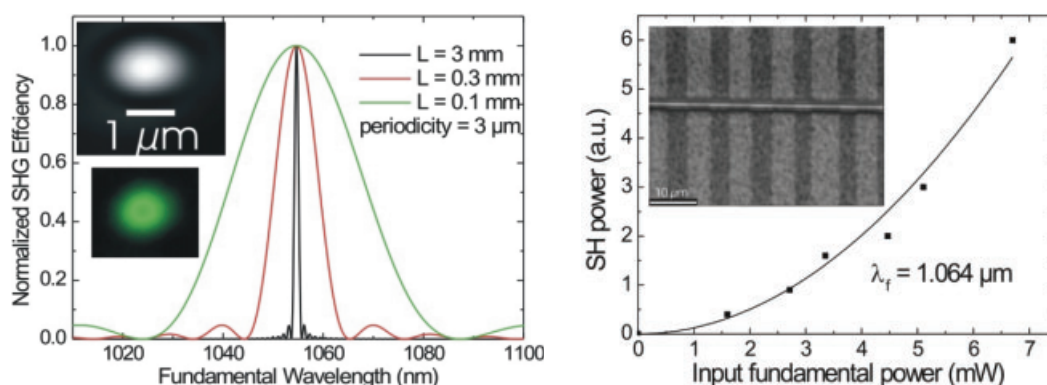


Figure 20 (online color at: www.lpr-journal.org) Left: Calculated (first order) quasi phase matching characteristics for SHG with different (interaction) lengths L . Upper and lower insets: photographs of TM-polarized modes of the fundamental ($\lambda = 1064$ nm) and the SH mode ($\lambda = 532$ nm), respectively. Right: measured (dots) and theoretical (fitted parabola, solid line) SH power versus input fundamental power (measured in front of the coupling lens). Inset: optical micrograph of a PPLN photonic wire of $1\ \mu\text{m}$ top width and $9.0\ \mu\text{m}$ periodicity.

interaction length, the broader the phase matching characteristic is. As only a section of $\sim 100\ \mu\text{m}$ length of the $3\ \text{mm}$ long waveguide was successfully poled, uncritical phase matching was expected.

To test SHG in the PPLNOI photonic wires, light of $1064\ \text{nm}$ wavelength from a narrow-band laser diode of up to $40\ \text{mW}$ output power was coupled to a waveguide. The SH output power was detected by a photomultiplier which is sensitive at the SH and blind at the fundamental wavelength. The insets of Fig. 20 (left) show the observed modes of a photonic wire of $1\ \mu\text{m}$ top width and $710\ \text{nm}$ thickness at $1064\ \text{nm}$ and $532\ \text{nm}$ wavelength, respectively. Due to the broad phase matching characteristics, SHG could also be observed in waveguides of $2\text{--}7\ \mu\text{m}$ width. The dependence of the generated SH power on the input fundamental power is shown in Fig. 20 (right); as expected, a parabolic dependence is observed.

5. Microring Resonators

Optical microring resonators are considered to be one of the main building blocks in future very-large-scale integrated photonic devices. We have realized two different types of LiNbO_3 microring resonators. Microrings of the first type are obtained by planar structuring of BCB-bonded LNOI samples and are thus suitable for high-density integration on a single chip. The second types are free-standing LiNbO_3 microrings which can be transferred onto any arbitrary host substrate to build hybrid integrated optics devices.

5.1. Microring resonators in LiNbO_3 thin films

Fig. 21 shows a schematic cross-section of the first electro-optically tunable microring resonators realized in Z-cut LiNbO_3 thin films [16]. The microrings are covered with a SiO_2 cladding and upper Cr electrodes. This configuration allows the applied electric field to be along the Z-axis of

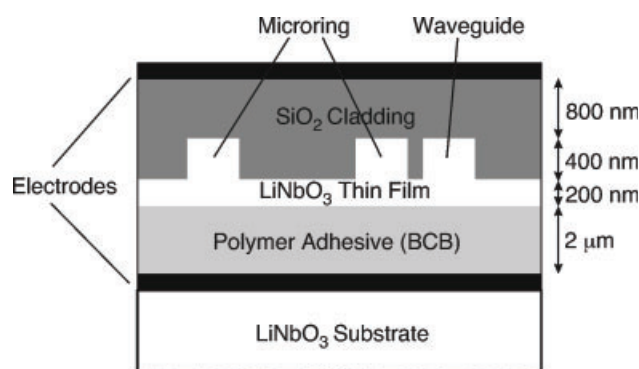


Figure 21 Schematic cross-section of an electro-optically tunable microring resonator based on BCB-bonded LNOI.

LiNbO_3 and therefore exploits the largest electro-optic coefficient r_{33} . It also enables to realize complex devices with separate upper electrodes for the independent control of different electro-optical microring resonators on a single chip.

Initially, the quality of these microring resonators was limited mainly by the lithography process performed by a standard mask aligner. Therefore, the minimum radius of useful microrings was limited to $100\ \mu\text{m}$, yielding a free spectral range (FSR) of $1.7\ \text{nm}$. Figure 22 shows an example of a transmission spectrum for TM polarized light and a shift of the resonance dip at $1555\ \text{nm}$ upon applying $100\ \text{VDC}$ on the electrodes. A shift of $0.11\ \text{nm}$ corresponds to a tunability of $0.14\ \text{GHz/V}$. An increase of the tunability is feasible by reducing the thickness of cladding layers (SiO_2 and BCB) and by using semi-transparent electrodes.

Using the recently developed laser lithography system [34], optical waveguides with better surface quality were obtained, allowing smaller curvature radii. Figure 23 shows a scanning electron micrograph of a racetrack microring resonator with a curvature radius of $20\ \mu\text{m}$ and a $10\text{-}\mu\text{m}$ long coupling region with a gap of $200\ \text{nm}$. From the transmission measurements a FSR of $7\ \text{nm}$ and a quality factor Q of around $10,000$ were deduced.

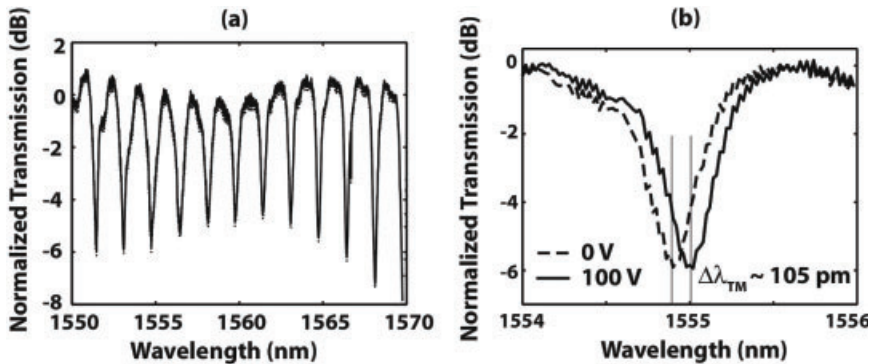


Figure 22 (a) Measured transmission (TM polarization) at the through port of a LiNbO₃ microring with a radius of 100 μm structured using a mask aligner. (b) Resonance dip at a wavelength of 1555 nm (dashed line) and the corresponding electro-optically shifted curve (solid line) upon applying 100 VDC to the device electrodes.

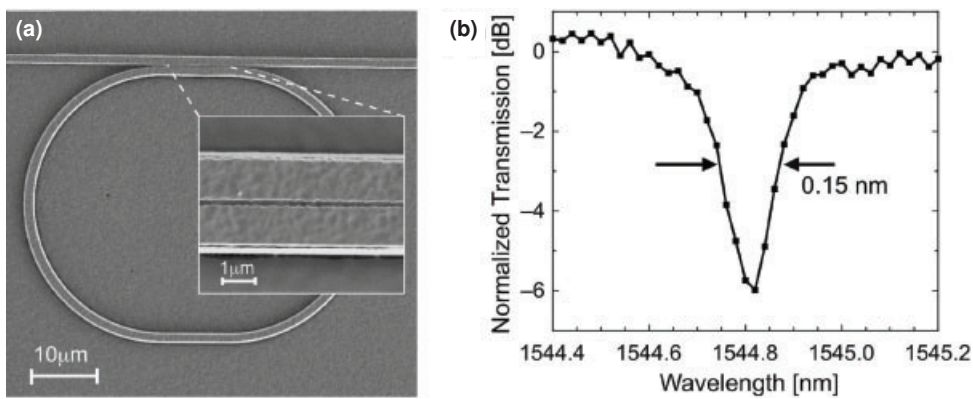


Figure 23 (a) Racetrack microring resonator in a LiNbO₃ thin film structured by laser lithography patterning and Ar⁺ ion etching. (b) Transmission resonance dip with a width of 0.15 nm (FWHM) at a wavelength of 1544.8 nm

5.2. Free-standing microrings for hybrid integration

Further design and functional versatility of microring resonators can be achieved by combining different materials for coupling waveguides and microrings, such as semiconductors, glass-compounds, polymers and organic epitaxial layers. For this purposes we developed a novel technique, which enables the fabrication of free-standing microrings from BCB-bonded LNOI. The structuring procedure is the same as described above; however, the rings are etched completely down to the BCB layer. A sample with isolated microrings is then put into a bath of standard clean-1 (SC-1) solution at a temperature of 100 °C. In this step, the adhesion between the LiNbO₃ microrings and the BCB layer is strongly reduced. The microrings remaining on the top of the substrate become freely movable and can be transferred onto any arbitrary host substrate where they can be positioned with submicrometer accuracy using micropositioning tools or other alignment techniques.

Based on this approach we realized for the first time LiNbO₃ microrings vertically coupled to gallium nitride (GaN) coupling waveguides on a sapphire substrate [45]. GaN exhibits low absorption in the near infrared wavelength range and a refractive index matching that of LiNbO₃. Figure 24 shows a scanning electron micrograph of a free-standing LiNbO₃ microresonator with a radius of 20 μm, which was positioned on a GaN waveguide.

In order to optimize the vertical light coupling, a sapphire substrate with GaN waveguides was first covered with a 100-nm thick layer of silicon nitride (PECVD at 300 °C).

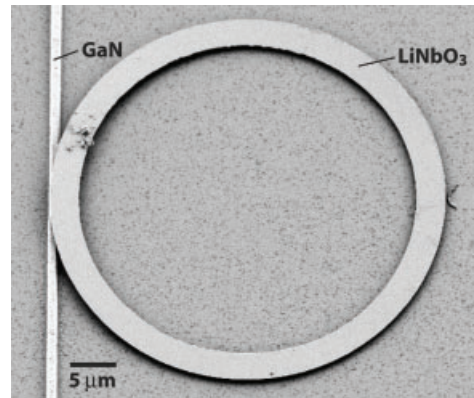


Figure 24 Scanning electron micrograph of a free-standing LiNbO₃ microring with a radius of 20 μm, positioned on a straight GaN waveguide. Below is a sapphire wafer used as a substrate. The widths of the coupling waveguide and of the microring are 1.1 μm and 3.0 μm, respectively. Both structures have a height of 600 nm.

Subsequently, free-standing LiNbO₃ microrings were transferred and aligned on top of the GaN waveguides. Finally, a thin cladding layer of SiN was deposited to fix the microrings at the aligned positions. A TE-polarized transmission spectrum of such a microring resonator is shown in Fig. 25. It exhibits sharp resonance dips with a linewidth of 0.15 nm FWHM and an extinction ratio of up to 18 dB, separated by a FSR = 7.9 nm at 1.55 μm.

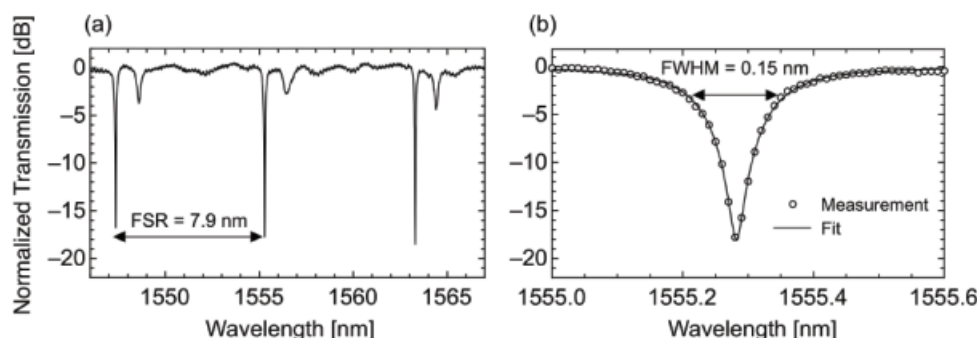


Figure 25 (a) Measured transmission spectrum (TE-polarization) through a GaN port waveguide of a free-standing LiNbO₃ microring resonator. (b) Detail of the measured resonance dip at 1555.28 nm with a linewidth of 0.15 nm FWHM and an extinction ratio of 18 dB.

From these measured values we deduced a finesse of 53 and quality factor $Q \sim 10,000$. From the fit of the measured resonance dip shown in Fig. 25b we estimated the total propagation losses of 17 dB/cm for the zero-order TE mode in the LiNbO₃ microring. The rather wide microring (2.5 μm) obviously supported also higher order modes as well, which are manifested in broader shallow dips between the sharp resonances.

6. Photonic Crystal Slabs

Photonic crystal (PhC) structures have a great potential to reduce the size and increase the functionality of integrated optics devices. Up to now, a remarkable progress in the realization of high quality PhC structures has been achieved mainly in silicon and some other semiconductor materials. PhC structures in LiNbO₃ are of great interest because they offer a possibility to build very small electro-optical modulators and switches. In the past, most of the studies on PhCs in LiNbO₃ have been performed using annealed proton exchanged waveguides with weak optical guiding [46–49]. However, the development of high-quality PhC structures requires single-crystalline LiNbO₃ slabs, which provide strong index guiding in the vertical direction. In the following we describe first PhC structures realized in BCB-bonded LNOI samples [30].

Using the FIB milling as described in Sect. 3.4, we fabricated various triangular photonic crystal lattices with different number of holes in the middle of ridge optical waveguides to facilitate the in- and out-coupling of probe laser light. Figure 26 shows an example of a ridge optical waveguide with a tapered section containing a triangular array of 15×36 air holes (lattice period = 500 nm, hole diameter = 240 nm). The tapered waveguide structure was chosen to recollect the diffracted light passing through the photonic crystal lattice. The fabricated sample with a length of 3 mm contained several parallel optical waveguides with and without photonic crystal structures for comparative transmission measurements. The total milling time for the shown array of holes amounted to around 6 min. Thus, focused ion beam milling is a very suitable method for fast prototyping of high-quality photonic crystal structures in LNOI.

In order to experimentally demonstrate the existence of photonic bandgaps in the fabricated structures we performed optical transmission measurements. An optical setup for the

transmission measurements is shown in Fig. 27. It consists mainly of a white light laser source, beam focusing and collimating optics, and a spectrometer (ANDO AQ-6315A) to analyze the transmitted spectrum. The laser source was a commercially available supercontinuum fiber laser (Fianium SC450) delivering 2W of average output power within a broad spectral range between 450 and 1950 nm. Two rotatable Glan-Taylor polarizers were used to adjust the polarization and power of the in-coupled light.

Fig. 28 depicts a measured normalized transmission spectrum (full line) of the triangular lattice shown in Fig. 26. The measurements were performed with TE-polarized light

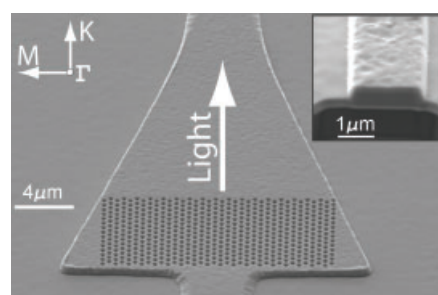


Figure 26 Scanning electron micrograph of a ridge optical waveguide with a tapered region containing a triangular photonic crystal lattice of 15×36 air holes. Light is propagating along symmetry direction $\Gamma \rightarrow K$. The inset shows an input facet of the coupling optical waveguide smoothed by FIB milling.

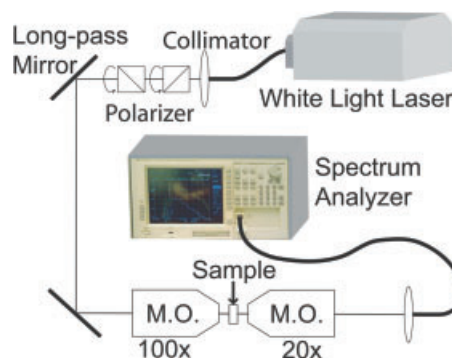


Figure 27 (online color at: www.lpr-journal.org) Optical set-up with a white light laser source and a spectrum analyzer for the characterization of the fabricated photonic crystal structures.

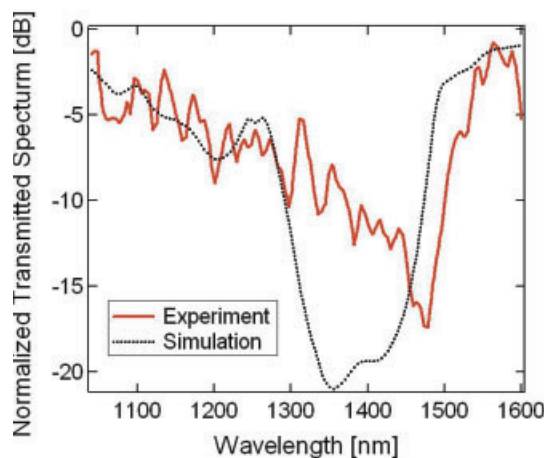


Figure 28 (online color at: www.lpr-journal.org) Measured (solid line) and calculated (dotted line) normalized transmission through the photonic crystal structure shown in Fig. 26.

(polarization parallel to the LNOI surface) over the wavelength range from 1050 to 1600 nm. The normalized transmission spectrum was obtained by dividing two transmission spectra of adjacent waveguides with and without a photonic crystal structure in the middle. As it can be seen, the spectrum shows a broad dip between 1300 and 1500 nm with the maximum extinction ratio of 15 dB at a wavelength of 1470 nm. In contrast to the TE polarization, no photonic gap was observed for TM polarized light, which is in accordance with a theoretical analysis of the fabricated structure. Although the fabricated slabs are not symmetric in the vertical direction (air/ LiNbO₃/BCB), the band gap effects are still possible due to strong confinement of propagation modes in the vertical direction [50]. The transmission properties of the fabricated structure were also modeled using a three-dimensional finite-difference time-domain software (OptiFDTD, Optiwave Systems Inc.). Figure 28 shows a comparison of the calculated transmission (dotted line) and the measured transmission (solid line). We contribute the deviations between both curves to differences between the designed (ideal) and fabricated structure.

7. Conclusions and Perspectives

The recent development of high-index contrast “lithium niobate on insulator” (LNOI) has been reviewed. Sub-micrometer thin single-crystalline LiNbO₃ films, bonded to a low-index substrate, are obtained by “ion-slicing” with He⁺-ions of ~200 keV energy. Two complementary techniques for bonding have been developed. Benzocyclobutene (BCB), which simultaneously serves as low-index cladding layer, enables relatively easy bonding to a LiNbO₃ substrate. On the other hand, crystal bonding – exploiting van der Waals forces – on a vacuum deposited SiO₂-layer on a LiNbO₃ substrate allows high annealing temperatures, important to reduce ion-implantation induced material damage and to recover electro- and nonlinear optical properties of the LiNbO₃ waveguide layer. A full wafer (3" diameter)

technology has been developed to fabricate LNOI of different material and layer compositions. Even periodically poled LNOI could be fabricated in this way.

Similar to SOI (silicon-on-insulator) LNOI might become a new platform for high-density integrated optics. It has – in contrast to SOI – excellent electro-optic, acousto-optic, and nonlinear optical properties. Moreover, it can be easily doped with rare-earth ions to get a laser active material. Therefore, several micro- and nano-structuring techniques, such as conventional and laser beam lithography of high resolution, chemical etching and (focused) ion-beam milling and focused ion-beam etching, have been used to develop a number of micro-photonic devices.

As basic structures, LNOI photonic wires with cross-section < 1 μm² have been developed. Their most interesting dispersion properties can be adjusted by the waveguide dimensions alone in a wide range. Also periodically poled (PP)LNOI photonic wires have been fabricated with a domain period of 9 μm and second harmonic generation (SHG) could be successfully demonstrated. Due to their small mode size, PPLNOI photonic wires have a great potential for the development of second order nonlinear devices of unprecedented efficiency. They are ideal candidates for the development of a wide range of nonlinear wavelength converters and of devices for all-optical signal processing. In particular, sub-micrometer ferroelectric domain structures would be very attractive enabling devices with backward nonlinear coupling.

Using a high-resolution laser lithography system, electro-optically tunable LNOI microring resonators with a waveguide curvature of 20 μm and a footprint of 45 × 55 μm² only have been developed. Their high quality is demonstrated by a *Q*-value up to 10,000. A tunability of 0.14 GHz/V could be shown. Moreover, free standing LiNbO₃ micro-rings of 20 μm radius have been developed for hybrid integration with nearly arbitrary materials; this is demonstrated by vertical coupling to a GaN channel guide on a sapphire substrate. A finesse of 53 and a free spectral range of 7.9 nm at 1.55 μm wavelength have been obtained.

Photonic crystal structures have been fabricated based on BCB-bonded LNOI. Both, simulation and experimental results show a photonic bandgap around 1470 nm wavelength, a first step towards the realization of electro-optically tunable photonic bandgap structures.

The growing activities and the great potential of LNOI based devices will lead to novel concepts and architectures of a high-density integrated optics with highly efficient electro-optical, nonlinear-optical, and laser/amplifier devices.

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